



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2010

**TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION**

Mathematics

General Instructions

- Reading time – 5 minutes.
- Working time – 180 minutes.
- Write using black or blue pen.
Pencil may be used for diagrams.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Start each **NEW** question in a separate answer booklet.

Total Marks - 120 Marks

- Attempt questions 1 – 10
- All questions are of equal value.

Examiner: *E. Choy*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Total marks – 120

Attempt Questions 1 – 10

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

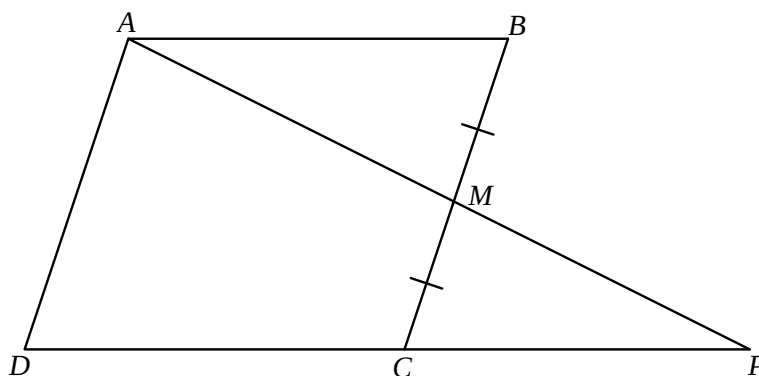
Question 1 (12 marks)	Use a SEPARATE writing booklet	Marks
(a) Evaluate $\frac{\sqrt{a^2 + b^2}}{c}$, if $a = 1.23$, $b = 0.8$ and $c = 4.81$. Leave your answer correct to 2 decimal places.		1
(b) Factorise $3m^2 - 13m + 4$		1
(c) If $\frac{5}{2 + \sqrt{3}} = a + b\sqrt{3}$, for rational a and b , by rationalising the denominator find a and b .		1
(d) Solve $ 2x - 1 > 5$ and graph the solution on a number line		2
(e) Solve the following equations simultaneously $3x + y = 6$ $6x - 2y = -8$		2
(f) Find a primitive of $5 + \sin x$.		2
(g) Express $\frac{3x-1}{4} - \frac{x-2}{3}$ as a single fraction in its simplest form.		1
(h) Given $\log_a 3 = 0.6$ and $\log_a 2 = 0.4$, find $\log_a 18$.		2

Question 2 (12 marks)

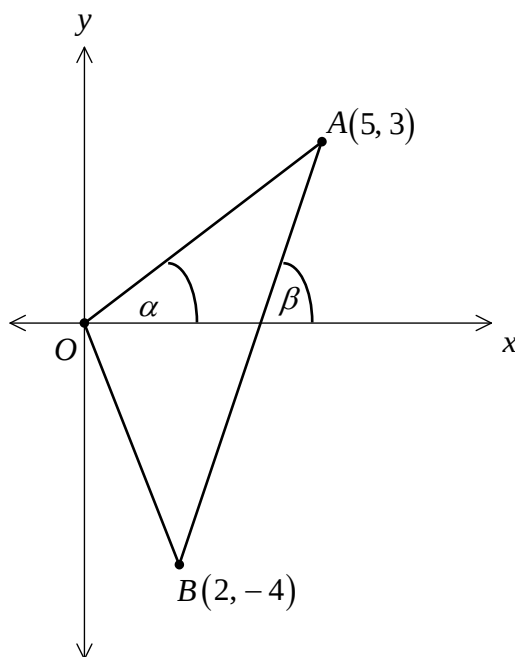
Use a SEPARATE writing booklet

Marks

- (a) The diagram below shows the parallelogram $ABCD$ with M the midpoint of BC . The intervals AM and DC are produced to meet at P .



- (i) Prove that $\triangle ABM \equiv \triangle PCM$ 2
- (ii) Hence prove that $ABPC$ is a parallelogram. 2
- (b) The diagram below shows $\triangle AOB$ with A and B the points $(5, 3)$ and $(2, -4)$ respectively. The angle of inclination of OA is α and the angle of inclination of AB is β .



- (i) Write down the gradients of OA and AB . 2
- (ii) Hence, find α and β , both correct to the nearest degree. 2
- (iii) Find the length of OA 1
- (iv) Find the length of AB . 1
- (v) Find the area of $\triangle AOB$. 2
Give your answer correct to two significant figures.

Question 3 (12 marks)

Use a SEPARATE writing booklet

Marks(a) Differentiate with respect to x

(i) $(e^x - 2)^5$ 2

(ii) $\frac{x^3}{\tan x}$ 2

(b) Find $\int \frac{3x}{x^2 - 1} dx$ 2

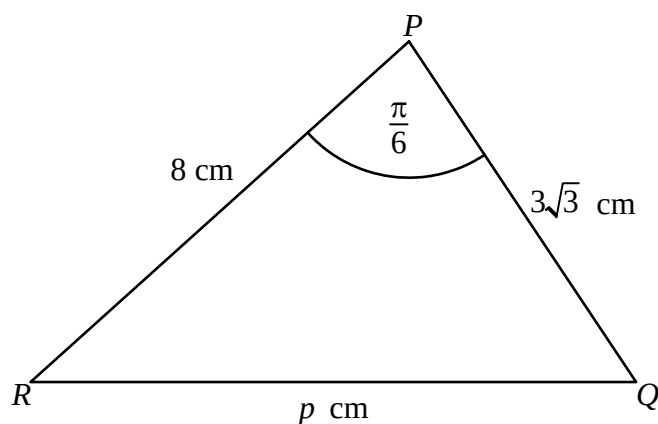
(c) Evaluate $\int_0^2 e^{-x} dx$. 2

Leave your answer in *exact* form.

(d) Solve $\tan x = -\sqrt{3}$ for $0 \leq x \leq 2\pi$ 1

(e) Sketch the graph of $x^2 + y^2 = 7$, showing all intercepts. 1

(f) In $\triangle PQR$ below, $\angle PRQ = \frac{\pi}{6}$, $PR = 8$ cm, $PQ = 3\sqrt{3}$ cm and $RQ = p$ cm.

Find the value of p in exact form. 1

(g) Given that $\sin \theta = \frac{3}{4}$ and $\tan \theta < 0$, find the exact value of $\cos \theta$. 1

Question 4 (12 marks)

Use a SEPARATE writing booklet

Marks

(a) Find the equation of the normal to $y = \log_e(3x - 2)$ at the point $(1, 0)$ 2

(b) Consider the quadratic equation $x^2 - kx + k + 3 = 0$, for k real.

(i) Find the discriminant and write it in simplest form. 1

(ii) For what values of k does the quadratic equation have no real roots. 1

(iii) If the product of the roots is equal to three times the sum of the roots, find the value of k . 1

(c) The table below shows the values of the function $f(x)$ for five values of x .

x	4	4.5	5	5.5	6
$f(x)$	1.3	2.9	0.7	-0.2	-1.1

Use Simpson's rule with these five function values to find an estimate for

2

$$\int_4^6 f(x) dx.$$

Give your answer correct to one decimal place.

(d) The equation of a parabola is given by $(x - 1)^2 = 8y$

(i) Write down the coordinates of the vertex. 1

(ii) Write down the focal length 1

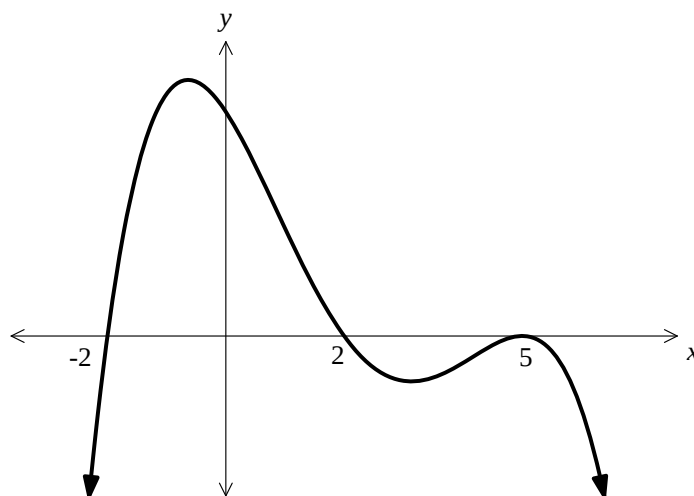
(iii) Sketch the graph of the parabola, clearly showing the focus and directrix. 1

(e) An infinite geometric series has a limiting sum of 24.

2

If the first term is 15, find the common ratio.

- (a) The graph of $y = f'(x)$ is shown below.



Sketch the graph of $y = f(x)$, given that $f(0) = 0$ and $f(5) = -3$.
Show clearly any turning points or points of inflexion.

3

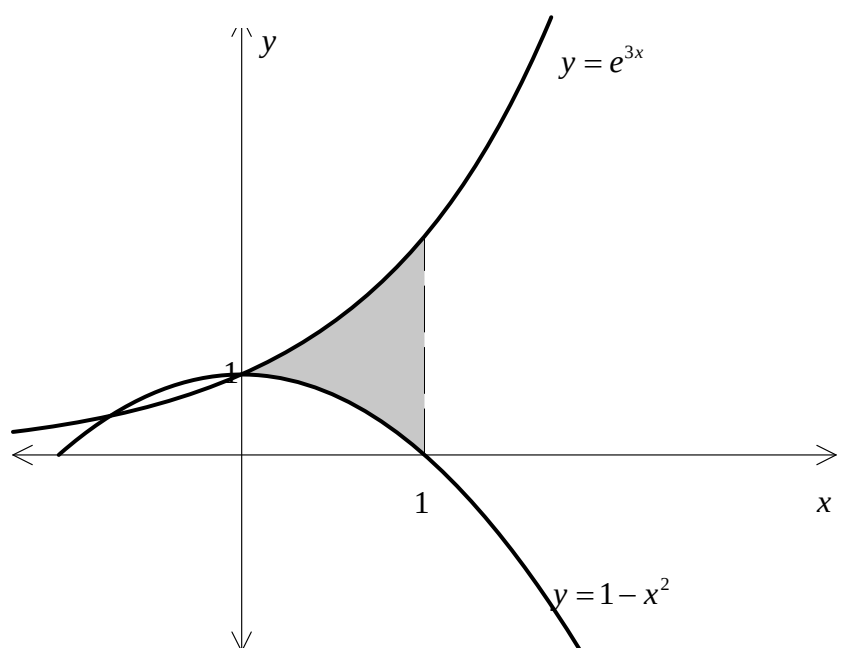
- (b) Differentiate $\log_e(\cos x)$ and express your answer in simplest form

2

- (c) Solve the equation $1 + \log_2 x = \log_2 \sqrt{x}$

2

- (d) The diagram below shows the region enclosed between the two curves, $y = e^{3x}$ and $y = 1 - x^2$, and the line $x = 1$.



Find the area of the shaded region.

3

- (e) Sketch the graph of the function $y = 2 \tan x$ for $0 \leq x \leq \frac{\pi}{2}$.
State the range.

2

Question 6 (12 marks)

Use a SEPARATE writing booklet

Marks

- (a) A particle P is moving in a straight line so that its velocity v metres per second after t seconds is given by $v = 12 - 4t$.

Initially, P is 3 metres to the right of the origin O .

- | | | |
|-------|---|----------|
| (i) | Find the initial velocity and acceleration of P . | 1 |
| (ii) | If the displacement of P from O is x metres, find an expression for x in terms of t . | 1 |
| (iii) | Find when and where P is at rest. | 1 |
| (iv) | Sketch the graph of $v = 12 - 4t$ for $0 \leq t \leq 5$. | 1 |
| (v) | Hence, or otherwise, find the total distance travelled by P during the first 5 seconds. | 2 |

- (b) A function is defined by $f(x) = \frac{x^3}{4}(x - 8)$

- | | | |
|-------|---|----------|
| (i) | Find the coordinates of the stationary points of the graph of $y = f(x)$ and determine their nature. | 2 |
| (ii) | Sketch the graph of $y = f(x)$ showing all its essential features including stationary points and intercepts. | 3 |
| (iii) | For what values of x is the curve increasing? | 1 |

Question 7 (12 marks)

Use a SEPARATE writing booklet

Marks

(a) The function $f(x) = e^x + e^{-x}$ is defined for all real values of x .

(i) Show that $f(x) = e^x + e^{-x}$ is an even function. **1**

(ii) Find the stationary point and its nature. **3**
Hence sketch the graph of $y = f(x)$.

(iii) The region bounded by the curve $y = e^x + e^{-x}$, the x -axis and the line $x = -2$ and $x = 2$ is rotated about the x -axis. **3**
Find the volume of the solid of revolution, correct to one decimal place.

(b) The population N of a certain species at time t is given by $N = N_0 e^{-0.03t}$, where t is in days and N_0 is the initial population of the species.

(i) Show that $N = N_0 e^{-0.03t}$ is a solution of the differential equation **1**

$$\frac{dN}{dt} = -0.03N$$

(ii) How long, to the nearest day, will it take for the population to halve? **1**

(iii) Find, in terms of N_0 , the rate of change of the population at the time when the population has halved. **1**

(iv) Find the number of days, to the nearest whole number, for the species' population to fall just below 5% of the initial number present **2**

Question 8 (12 marks)

Use a SEPARATE writing booklet

Marks

- (a) A couple plan to buy a home and they wish to save a deposit of \$40 000 over five years. They agree to invest a fixed amount of money at the beginning of each month during this time. Interest is calculated at 12% per annum compounded monthly.

- (i) Let P be the monthly investment. Show that the total investment A after five years is given by

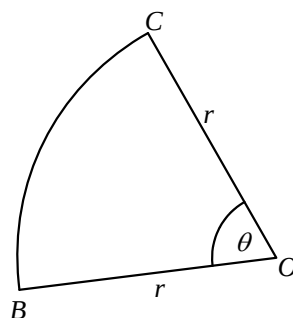
2

$$A = P(1 \cdot 01 + 1 \cdot 01^2 + \dots + 1 \cdot 01^{60})$$

- (ii) Find the amount P needed to be deposited each month to reach their goal. Answer correct to the nearest dollar.

2

- (b) The diagram below shows a sector OBC of a circle with centre O and radius r cm. The arc BC subtends an angle θ radians at O .



- (i) Show that the perimeter of the sector is $r(2 + \theta)$

2

- (ii) Given that the perimeter of the sector is 36 cm, show that its area is given by

2

$$A = \frac{648\theta}{(\theta + 2)^2}$$

- (iii) Hence show that the maximum area of the sector is 81 cm^2

4

- (a) An underground storage tank is in the shape of a rectangular prism with a floor area of 12 m^2 and a ceiling height of 2 m.

At 2 p.m. one Sunday, rain water begins to enter the storage tank.

The rate at which the volume V of the water changes over time t hours is given by

$$\frac{dV}{dt} = \frac{24t}{t^2 + 15}$$

where $t = 0$ represents 2 p.m. on Sunday and where V is measured in cubic metres. The storage tank is initially empty.

- (i) Show that the volume of water in the tank at time t is given by 3

$$V = 12 \log_e \left(\frac{t^2 + 15}{15} \right), \quad t \geq 0$$

- (ii) Find the time when the tank will be completely filled with water if the water continues to enter the tank at the given rate. 3
Express your answer to the nearest minute.

- (iii) The owners return to the house and manage to simultaneously stop the water entering the tank and start the pump in the tank. 3
This occurs at 6 p.m. on Sunday.
The rate at which the water is pumped out of the tank is given by

$$\frac{dV}{dt} = \frac{t^2}{k} \quad \text{where } k \text{ is a constant}$$

At exactly 8 p.m. the tank is emptied of water.

Find the value of k .

Express your answer correct to 4 significant figures.

- (b) The captain of the submarine, the HMAS Yddap, spots a freighter on the horizon. He knows that a single torpedo has a probability of $\frac{1}{4}$ of sinking the freighter, $\frac{1}{2}$ of damaging it and $\frac{1}{4}$ of missing it.
He also knows that 2 damaging shots will sink the freighter.
If two torpedoes are fired independently, find the probability of

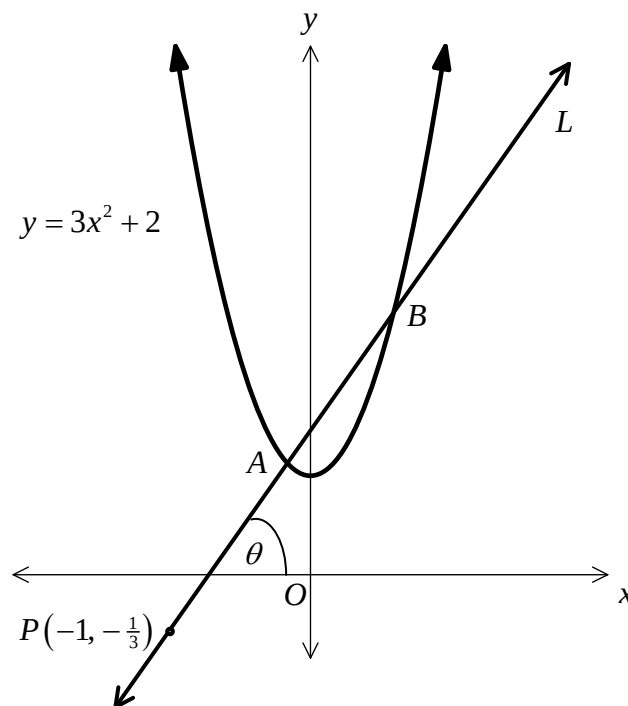
- (i) sinking the freighter with 2 damaging shots; 1
(ii) sinking the freighter. 2

Let L be the straight line passing through $P(-1, -\frac{1}{3})$ with angle of inclination θ to the x -axis. It is known that the coordinates of any point Q on L are in the form $(-1 + r \cos \theta, -\frac{1}{3} + r \sin \theta)$, where r is a real number.

(a) Show that $PQ = |r|$.

2

(b) In the figure below, L cuts the parabola $y = 3x^2 + 2$ at point A and B . Let $PA = r_1$ and $PB = r_2$.



(i) By considering the fact that the points A and B lie both on the line L and the parabola $y = 3x^2 + 2$, show that r_1 and r_2 are the roots of the equation

2

$$9r^2 \cos^2 \theta - 3r(\sin \theta + 6 \cos \theta) + 16 = 0$$

(ii) Using b (i), show that $AB^2 = \frac{(\sin \theta - 2 \cos \theta)(\sin \theta + 14 \cos \theta)}{9 \cos^4 \theta}$.

3

(iii) Let L_1 be a tangent to the parabola $y = 3x^2 + 2$ from P , with point of contact R . Using the above results, find the two possible slopes of L_1 .

2

(iv) Show that $PR = \frac{4\sqrt{5}}{3}$ when one of the slopes of L_1 has a value of 2.

3

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Question 1

$$(a) \frac{\sqrt{1.23^2 + 0.8^2}}{4.81} \approx 0.31$$

$$(b) 3m^2 + 3m + 4 \quad \begin{matrix} 3m & \times & -1 \\ m & \times & -4 \end{matrix}$$

$$= (3m-1)(m-4)$$

$$(c) \frac{5}{2+\sqrt{3}} = \frac{5}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}}$$

$$= \frac{10-5\sqrt{3}}{4-3}$$

$$= 10-5\sqrt{3}$$

$$\therefore a = 10, b = -5$$

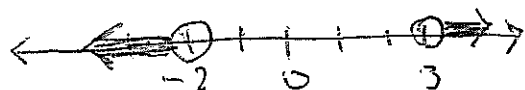
$$(d) |2x-1| > 5$$

$$-5 > 2x-1 \quad \text{or} \quad 2x-1 > 5$$

$$0 > 2x+4 \quad \quad \quad 2x > 6$$

$$-4 > 2x \quad \quad \quad x > 3$$

$$x < -2$$



$$(e) \begin{aligned} 3x + y &= 6 & \text{--- ①} \\ 6x - 2y &= -8 & \text{--- ②} \end{aligned}$$

$$2 \times \text{①}: 6x + 2y = 12 \quad \text{--- ③}$$

$$2 + \text{③}: 12x = 4$$

$$x = \frac{1}{3}$$

$$\text{in ①} \quad 1 + y = 6$$

$$y = 5$$

$$(f) \int 5 + \sin x \, dx = 5x - \cos x + C$$

$$(g) \frac{3x-1}{4} - \frac{x-2}{3}$$

$$= \frac{3(3x-1) - 4(x-2)}{12}$$

$$= \frac{9x-3-4x+8}{12}$$

$$= \frac{5x+5}{12}$$

$$(h) \log_a 3 = 0.6; \log_a 2 = 0.4$$

$$\log_a 18 = \log_a (3^2 \times 2)$$

$$= 2 \log_a 3 + \log_a 2$$

$$= 2 \times (0.6) + 0.4$$

$$= 1.2 + 0.4$$

$$= 1.6$$

Q2(a)

(i) $\angle BAM = \angle MPC$ (alt $\angle$ s $AB \parallel DP$)
 $\angle AMB = \angle CMP$ (vert. opp.)
 $BM = MC$ (given).
 $\therefore \triangle ABM \equiv \triangle PCM$ (AAS).

(ii) $AM = MP$ (corresponding sides congruent $\triangle$ s)
 $\therefore ABPC$ is a parallelogram since diagonals bisect each other.

(b) (i) $m_{OA} = \frac{5}{3}$ $m_{AB} = \frac{7}{3}$

(ii) $\alpha = 59^\circ 2'$ $\beta = 66^\circ 48'$

(iii) $OA = \sqrt{25+9}$
 $= \sqrt{34}$.

(iv) $AB^2 = (5-2)^2 + (3+4)^2$

$$AB^2 = 9 + 49$$

$$= 58$$

$$AB = \sqrt{58}$$

(v) $\angle OAB = \beta - \alpha$
 $= 7^\circ 46'$

$$\text{Area} = \frac{1}{2} \times \sqrt{34} \times \sqrt{58} \times \sin(\angle OAB)$$

$$= 3.$$

Question 3:

a) i) $(e^x - 2)^5$

$$\frac{d}{dx} = \underset{\textcircled{1}}{5(e^x - 2)^4} \times \underset{\textcircled{1}}{e^x}$$

$$= 5e^x (e^x - 2)^4$$

ii) $\frac{x^3}{\tan x} = \frac{u}{v}$

$$u = x^3 \quad v = \tan x \quad \textcircled{1}$$

$$u' = 3x^2 \quad v' = \sec^2 x$$

$$\frac{d}{dx} = \frac{\tan x \times 3x^2 - x^3 \times \sec^2 x}{\tan^2 x}$$

$$= \frac{3x^2 \tan x - x^3 \sec^2 x}{\tan^2 x} \quad \textcircled{1}$$

b) $\int \frac{3x}{x^2 - 1} dx$

$$= \frac{3}{2} \int \frac{2x}{x^2 - 1} dx \quad \textcircled{1}$$

$$= \frac{3}{2} \log_e (x^2 - 1) + C \quad \textcircled{1}$$

c) $\int_0^2 e^{-x} dx$

$$= [-e^{-x}]_0^2 \quad \textcircled{1}$$

$$= -e^{-2} - -e^0$$

$$= -e^{-2} + 1$$

$$= 1 - e^{-2} \quad \textcircled{1}$$

d) $\tan \alpha = -\sqrt{3}$

S/A
T/C

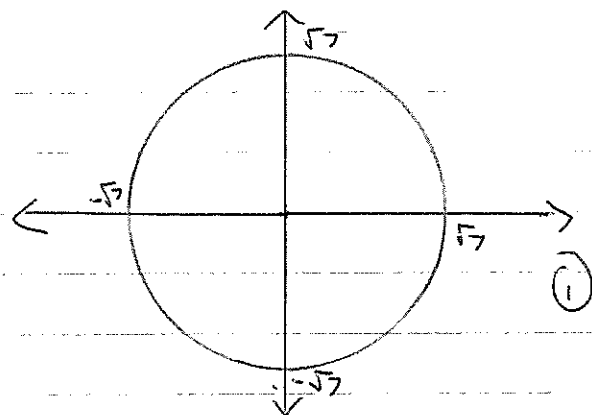
$$\therefore \alpha = -\pi/3$$

for $0 \leq \alpha \leq 2\pi$

$$\alpha = \frac{2\pi}{3}, \frac{5\pi}{3} \quad \textcircled{1}$$

e) $x^2 + y^2 = 7$

$\therefore$ circle radius $\sqrt{7}$



f) $p^2 = 8^2 + (3\sqrt{3})^2 - 2 \times 8 \times 3\sqrt{3} \times \cos \pi/6$

$$= 91 - (48\sqrt{3} \times \sqrt{3}/2)$$

$$= 91 - 72$$

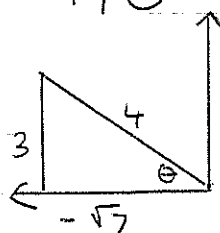
$$= 19$$

$$p = \sqrt{19} \text{ cm} \quad \textcircled{1}$$

g) $\sin \theta = 3/4$ $\tan < 0$

S/A
T/C

$$\therefore 90^\circ < \theta < 180^\circ$$



$$\cos \theta = -\frac{\sqrt{7}}{4} \quad \textcircled{1}$$

2010 MATHEMATICS (2U) TRIAL HSC.

QUESTION FOUR

a) $y = \ln(3x-2)$

gradient tangent =

$$\frac{dy}{dx} = \frac{3}{3x-2}$$

gradient normal =

$$-\frac{dx}{dy} = \frac{2-3x}{3}$$

at (1,0) $-\frac{dx}{dy} = \frac{-3+2}{3} = -\frac{1}{3}$

Eqn of normal

$$y - y_1 = \frac{dx}{dy}(x - x_1)$$

$$y - 0 = -\frac{1}{3}(x - 1)$$

$$3y = 1 - x$$

$$x + 3y - 1 = 0$$

b)(i) $x^2 - Rx + R + 3 = 0$

$$\Delta = b^2 - 4ac$$

$$= (-R)^2 - 4(1)(R+3)$$

$$= R^2 - 4R - 12$$

ii) For no real roots $\Delta < 0$

$$R^2 - 4R - 12 < 0$$

$$(x-6)(x+2) < 0$$

$$-2 < x < 6$$

b)(iii) Let the root be $\alpha + \beta$

$$\alpha\beta = \frac{c}{a} = R+3$$

$$\alpha + \beta = -\frac{b}{a} = -(-R) = R$$

NOW $\alpha\beta = 3(\alpha + \beta)$

$$R+3 = 3R$$

$$R = 3/2$$

c) Simpson's Rule

$$A \doteq \frac{h}{3} [y_1 + y_{n+1} + 4(y_{\text{odd}}) + 2(y_{\text{even}})]$$

$$h = \frac{b-a}{n} = \frac{6-4}{4} = \frac{1}{2}$$

$$A \doteq \frac{1}{6} [1 \cdot 3 - 1 \cdot 1 + 2(0.7) + 4(2.9 - 0.2)]$$

$$\doteq \frac{1}{6} [0.2 + 1.4 + 10.8]$$

$$\doteq 2.1 \text{ units}^2$$

$$\int_4^6 f(x) dx \doteq 2.1 \text{ units}^2$$

d) $(x-1)^2 = 8y$
 $(x-1)^2 = 4(2)y$

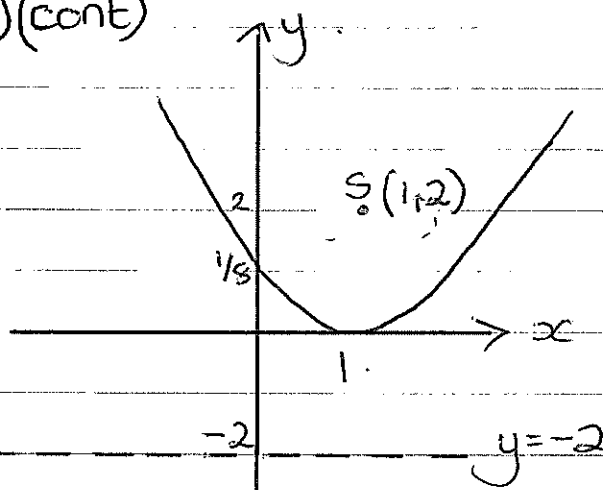
vertex = (1, 0)

Focal length = 2

2010 Mathematics (2U) Trial HSC.

Question 4

d)(cont)



e) limiting sum = 24

$$|r| < 1$$

$$\frac{a}{1-r} = 24$$

$$a = 15$$

$$\frac{15}{1-r} = 24$$

$$15 = 24 - 24r$$

$$24r = 9$$

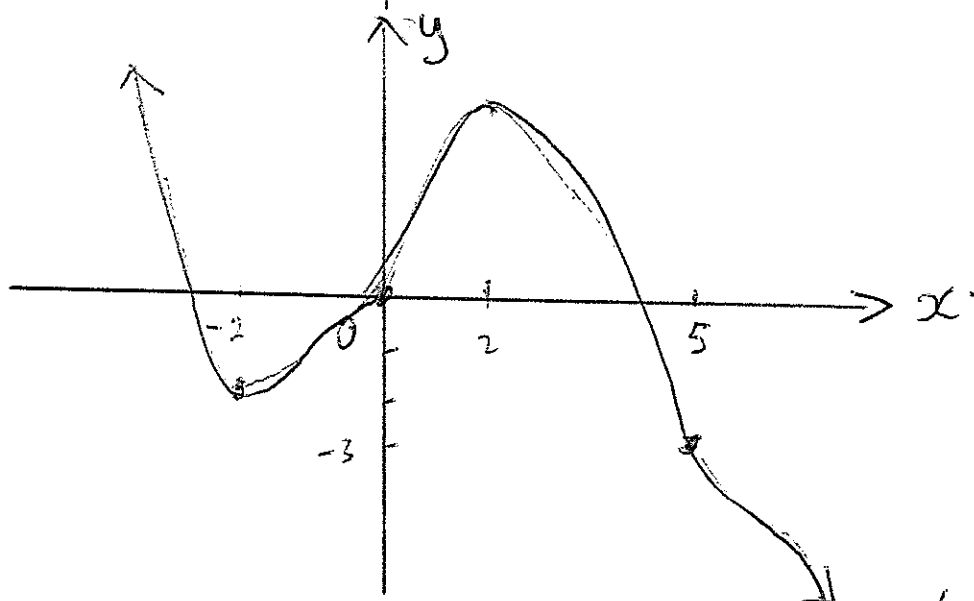
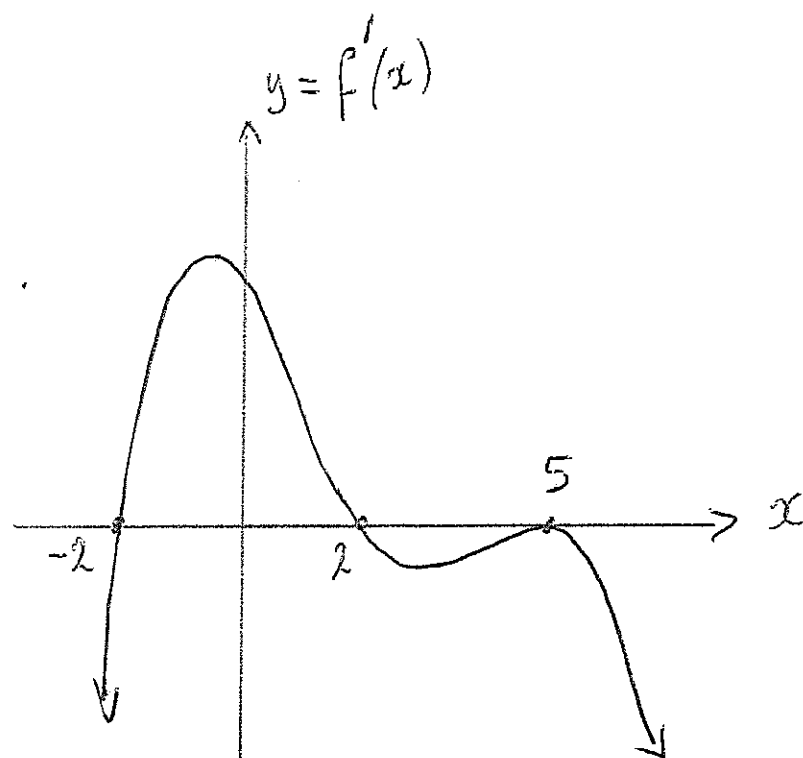
$$r = \frac{9}{24} = \frac{3}{8}$$

Q5

(a)

24

2019



At $x = -2$, $y' = 0$.

$$\left. \begin{array}{l} x = -2 - \varepsilon \quad y' < 0 \\ x = -2 + \varepsilon \quad y' > 0 \end{array} \right\} \begin{array}{l} \text{min} \\ \text{t. pt.} \end{array}$$

At $x = 2$, $y' = 0$

$$\left. \begin{array}{l} x = 2 - \varepsilon \quad y' > 0 \\ x = 2 + \varepsilon \quad y' < 0 \end{array} \right\} \begin{array}{l} \text{max} \\ \text{t. pt.} \end{array}$$

At $x = 5$, $y' = 0$

$$\left. \begin{array}{l} x = 5 - \varepsilon \quad y' < 0 \\ x = 5 + \varepsilon \quad y' < 0 \end{array} \right\} \begin{array}{l} \text{min / local} \\ \text{t. pt.} \end{array}$$

Also data $f(0) = 0$
 $f(5) = -3$.

(3)

$$\textcircled{5} \quad (b) \quad \frac{d}{dx} (\log_2 \cos x) = \frac{1}{\cos x} \times -\sin x$$

$$= \frac{-\sin x}{\cos x} = -\tan x \quad \textcircled{2}$$

$$(c) \quad 1 + \log_2 x = \log_2 \sqrt{x}$$

$$1 + \log_2 x = \frac{1}{2} \log_2 x$$

$$1 \log_2 x - \frac{1}{2} \log_2 x = -1$$

$$\frac{1}{2} \log_2 x = -1$$

$$\log_2 x = -2$$

$$\text{so } 2^{-2} = x$$

$$x = \frac{1}{4} \quad (0.25) \quad \textcircled{2}$$

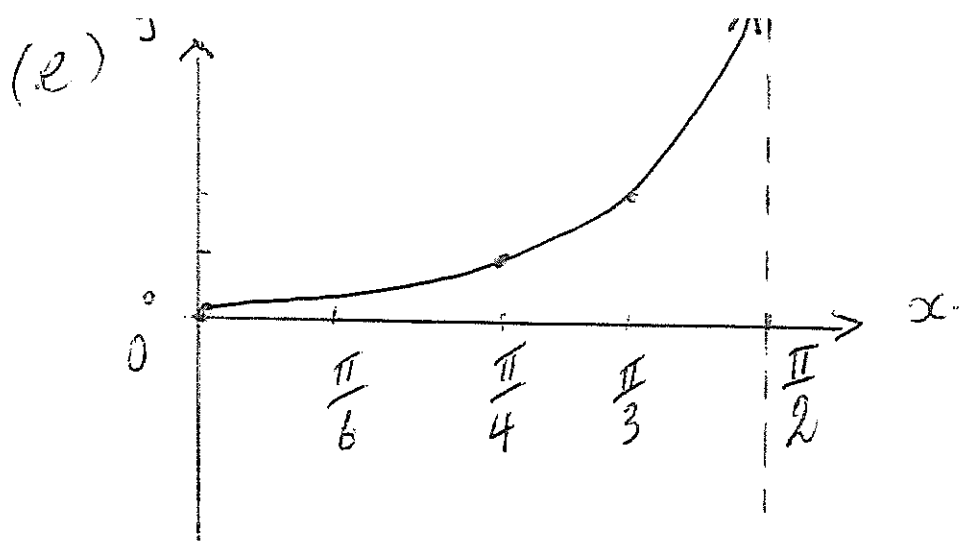
$$(d) \quad A = \int_0^1 (e^{3x} - (1-x^2)) dx$$

$$= \int_0^1 (e^{3x} - 1 + x^2) dx$$

$$= \left[\frac{1}{3} e^{3x} - x + \frac{x^3}{3} \right]_0^1$$

$$= \left(\frac{1}{3} e^3 - 1 + \frac{1}{3} \right) - \left(\frac{1}{3} e^0 - 0 + \frac{0}{3} \right)$$

$$= \frac{1}{3} e^3 - 1 + \frac{1}{3} - \frac{1}{3} = \frac{1}{3} e^3 - 1 \quad \text{units}^2 \quad \textcircled{3}$$



$$y = 2 \tan x.$$

①

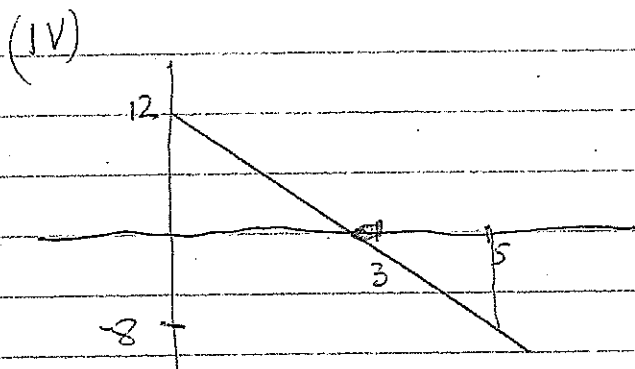
Range: $y \geq 0$ ①

b(a)

(i) $t = 0$ $v = 12$
 $a = \frac{dv}{dt} = -4$

(ii) $x = 12t - 2t^2 + c$ to $x=3$
 $c = 3$
 $x = 12t - 2t^2 + 3$

(iii) $v = 0$ $t = 3$
 $x = (12)(3) - 2(9) + 3$
 $x = 21$



$t = 0$ $x = 3$

$t = 3$ $x = 21$ 18 m travelled

$t = 5$ $x = 13$ 8 m travelled
 total 26 m travelled

or area under vel curve

$\frac{1}{2} \times 12 \times 3 + \frac{1}{2} \times 2 \times 8$

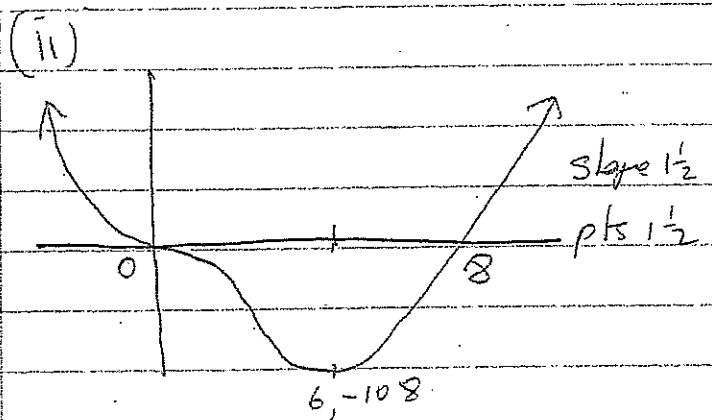
$= 26 \text{ m}$

b(b)

$f(x) = \frac{x^3}{4}(x-8) = \frac{x^4}{4} - 2x^3$

$f'(x) = x^3 - 6x^2 = x^2(x-6)$
 Stlpts $x = 0$ 6
 $y = -108$

$x = -1$ $f'(x) < 0$ $x = 1$ $f'(x) < 0$
 horizontal pt. / inflexion
 $x = 5$ $f'(x) < 0$ $x = 7$ $f'(x) > 0$
 min. turn. pt.



(iii)

increasing $x > 6$

Question 7

$$\begin{aligned} \text{a) i) } f(x) &= e^x + e^{-x} \\ f(-x) &= e^{-x} + e^{-(-x)} \\ &= e^x + e^{-x} \end{aligned}$$

$$\text{since } f(-x) = f(x)$$

$f(x)$ is an even function

$$\text{ii) } f'(x) = e^x - e^{-x}$$

$$\text{let } f'(x) = 0$$

$$e^x - \frac{1}{e^x} = 0$$

$$e^{2x} - 1 = 0$$

$$e^{2x} = 1$$

$$2x = \ln 1$$

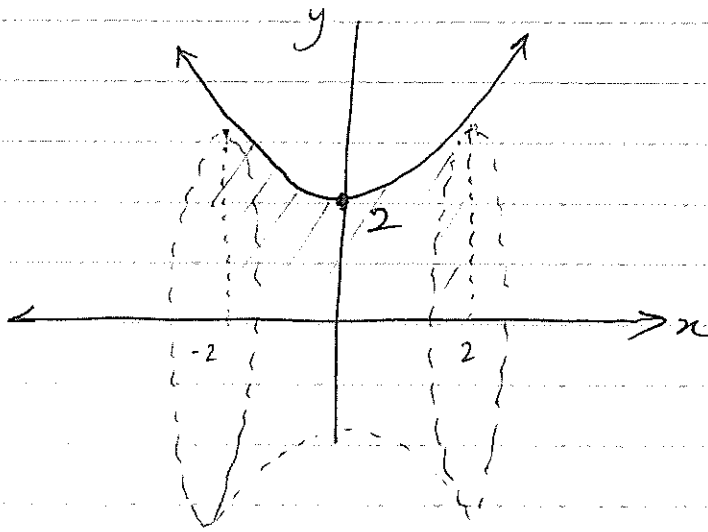
$$2x = 0$$

$$x = 0$$

$$\begin{aligned} f''(x) &= e^x + e^{-x} \\ &> 0 \text{ for all } x \end{aligned}$$

$$\begin{aligned} f(0) &= e^0 + e^{-0} \\ &= 2 \end{aligned}$$

$\therefore$ Minimum Turning Point at $(0, 2)$



$$\text{iii)} \quad V = \pi \int_a^b y^2 dx$$

$$V = \pi \int_{-2}^2 (e^x + e^{-x})^2 dx$$

$$V = 2\pi \int_0^2 (e^{2x} + 2 + e^{-2x}) dx \quad (\text{since even})$$

$$V = 2\pi \left[\frac{1}{2} e^{2x} + 2x - \frac{1}{2} e^{-2x} \right]_0^2$$

$$V = 2\pi \left[\frac{1}{2} e^{2(2)} + 2(2) - \frac{1}{2} e^{-2(2)} - \left(\frac{1}{2} e^{2(0)} + 2(0) - \frac{1}{2} e^{-2(0)} \right) \right]$$

$$V = 2\pi \left[\frac{1}{2} e^4 - \frac{1}{2} e^{-4} + 4 \right]$$

$$V = \pi [e^4 - e^{-4} + 8]$$

$$V \approx 196.6 \text{ units}^2$$

$$\text{b) i)} \quad N = N_0 e^{-0.03t}$$

$$\frac{dN}{dt} = -0.03 N_0 e^{-0.03t}$$

$$\frac{dN}{dt} = -0.03 N$$

$$\text{ii)} \quad \text{when } N = \frac{N_0}{2}$$

$$\frac{N_0}{2} = N_0 e^{-0.03t}$$

$$e^{-0.03t} = \frac{1}{2}$$

$$-0.03t = \ln\left(\frac{1}{2}\right)$$

$$t = \frac{\ln\left(\frac{1}{2}\right)}{-0.03}$$

$$t = 23 \text{ days (to nearest day)}$$

iii) when $N = \frac{N_0}{2}$

$$\frac{dN}{dt} = -0.03 \frac{N_0}{2}$$

$$\frac{dN}{dt} = -0.015 N_0$$

iv) when $N < 0.05 N_0$

$$N_0 e^{-0.03t} < 0.05 N_0$$

$$e^{-0.03t} < 0.05$$

$$\ln e^{-0.03t} < \ln 0.05$$

$$-0.03t < \ln 0.05$$

$$t > \frac{\ln 0.05}{-0.03}$$

$$t > 99.85$$

$$t = 100 \text{ days (to nearest whole number)}$$

QUESTION 8. (20)

(a) (i) $n = 60$ (ie 5 years)

$i\% = 0.1$ (interest per month)

$$\begin{aligned} \text{Total } A &= P(1.01)^{60} + P(1.01)^{59} + \dots + P(1.01)^1 \\ &= P(1.01 + 1.01^2 + 1.01^3 + \dots + 1.01^{60}) \end{aligned}$$

$$(ii) \quad 40\,000 = \frac{P(1.01(1.01^{60} - 1))}{1.01 - 1.0}$$

$$\begin{aligned} \therefore P &= \frac{40\,000 \times 0.01}{1.01(1.01^{60} - 1)} \\ &= \$485. \end{aligned}$$

(b) (i) $P = r + r + l$ (where $l = r\theta$)

$$\begin{aligned} &= r + r + r\theta \\ &= 2r + r\theta \\ &= r(2 + \theta) \end{aligned}$$

(ii) $A = \frac{1}{2} r^2 \theta$ where $36 = r(2 + \theta)$

$$= \frac{1}{2} \times \left(\frac{36}{2 + \theta}\right)^2 \times \theta \quad \therefore r = \frac{36}{2 + \theta}$$

$$= \frac{648\theta}{(2 + \theta)^2}$$

$$(iii) \quad A' = \frac{(2 + \theta)^2 \times 648 - 648\theta \times 2(2 + \theta)}{(2 + \theta)^4}$$

$$= \frac{648(2 + \theta) - 1296\theta}{(2 + \theta)^3}$$

$$= \frac{1296 - 648\theta}{(2 + \theta)^3}$$

$$\begin{aligned} \text{If } A' = 0 \quad \theta &= 2 \quad \& \quad A = \frac{648 \times 2}{16} \\ &= 81 \text{ cm}^2 \end{aligned}$$

Test

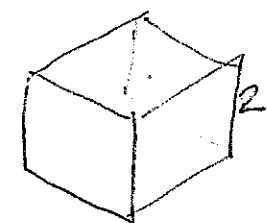
θ	1.5	2	2.5
A'	$\frac{324}{(3.5)^3} > 0$	0	$\frac{-324}{(4.5)^3} < 0$

$\therefore$ MAX.

QUESTION 9

(a)

$$\frac{dV}{dt} = \frac{24t}{t^2 + 15}$$


 $V = 0$ when $t = 0$

$$(i) \int \frac{dV}{dt} dt = 12 \int \frac{2t}{t^2 + 15} dt + C$$

$$V = 12 \ln(t^2 + 15) + C$$

Initially $0 = 12 \ln(15) + C$

$$\therefore C = -12 \ln(15)$$

$$* \therefore V = 12 \ln\left(\frac{t^2 + 15}{15}\right) \quad [2]$$

(ii) When full $V = 24$

$$24 = 12 \ln\left(\frac{t^2 + 15}{15}\right)$$

$$2 = \ln\left(\frac{t^2 + 15}{15}\right)$$

$$\therefore \frac{t^2 + 15}{15} = e^2$$

$$t^2 = 15e^2 - 15$$

$$t = \sqrt{15e^2 - 15}$$

$$\approx 9.7896 \text{ hrs}$$

$$[2\frac{1}{2}] \rightarrow = 9 \text{ hr } 47' 23''$$

Time 11:47 pm $[3]$

* Differentiate $\rightarrow [1\frac{1}{2}]$

(iii) At 6:00 $t = 4$ (filling)

$$V = 12 \ln\left(\frac{16 + 15}{15}\right)$$

$$\approx 8.7112 \text{ m}^3$$

Emptying

$$\frac{dV}{dt} = \frac{t^2}{k}$$

$$V = \frac{t^3}{3k} + C$$

At 6:00 pm, $t = 0$, $V = 0$ (pumped out)

$$\therefore C = 0$$

$$\text{So } V = \frac{t^3}{3k}$$

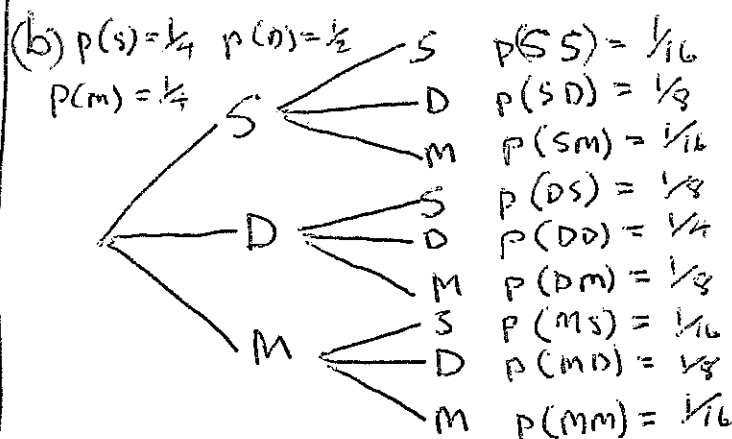
At 8:00 pm, $t = 2$, $V = 8.7112$
(All pumped out).

$$k = \frac{t^3}{3V}$$

$$= \frac{8}{3V}$$

$[3]$

$$\therefore k = 0.3061 \text{ or } 0.3061$$



$$(i) p(DD) = \frac{1}{4} \quad [2]$$

$$(ii) p(\text{Sink}) = p(S \times) + p(DS) + p(DD) + p(M \times)$$

$$= \frac{1}{4} + \frac{1}{8} + \frac{1}{4} + \frac{1}{16}$$

$$= \frac{11}{16} \quad [2]$$

2010 Mathematics Trial HSC: **Question 10** solutions

10. Let L be the straight line passing through $P(-1, -\frac{1}{3})$ with an angle of inclination θ to the x -axis. It is known that the coördinates of any point Q on L are in the form $(-1 + r \cos \theta, -\frac{1}{3} + r \sin \theta)$, where r is a real number.

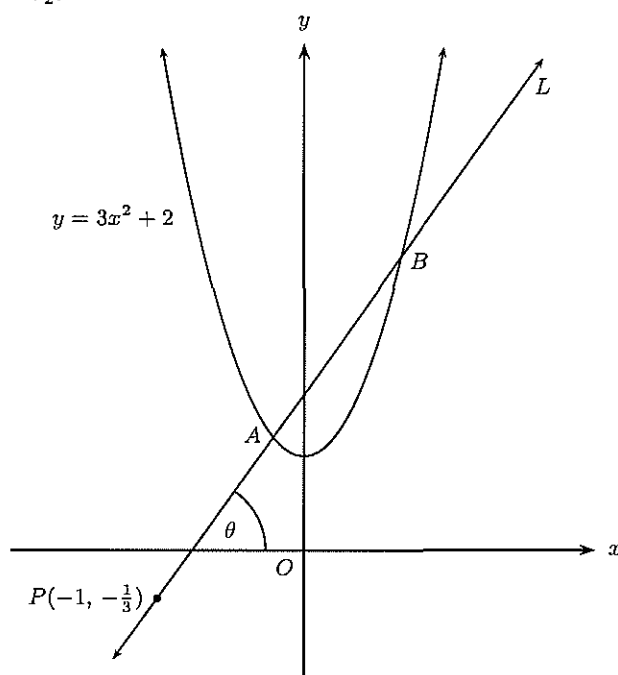
(a) Show that $PQ = |r|$.

1

Solution:

$$\begin{aligned} PQ &= \sqrt{(-1 + 1 - r \cos \theta)^2 + (-\frac{1}{3} + \frac{1}{3} - r \sin \theta)^2}, \\ &= \sqrt{r^2 \cos^2 \theta + r^2 \sin^2 \theta}, \\ &= \sqrt{r^2}. \\ \therefore PQ &= |r|. \end{aligned}$$

- (b) In the figure below, L cuts the parabola $y = 3x^2 + 2$ at the points A and B . Let $PA = r_1$ and $PB = r_2$.



- (i) By considering points A and B lie both on the line L and the parabola $y = 3x^2 + 2$, show that r_1 and r_2 are the roots of the equation

3

$$9r^2 \cos^2 \theta - 3r(\sin \theta + 6 \cos \theta) + 16 = 0.$$

Solution: Method 1—

Where Q cuts the parabola $y = 3x^2 + 2$:

$$\begin{aligned} -\frac{1}{3} + r \sin \theta &= 3(-1 + r \cos \theta)^2 + 2, \\ -1 + 3r \sin \theta &= 9(1 - 2r \cos \theta + r^2 \cos^2 \theta) + 6, \\ 3r \sin \theta &= 9 - 18r \cos \theta + 9r^2 \cos^2 \theta + 7, \\ 9r^2 \cos^2 \theta - 18r \cos \theta - 3r \sin \theta + 16 &= 0, \\ \text{i.e. } 9r^2 \cos^2 \theta - 3r(\sin \theta + 6 \cos \theta) + 16 &= 0. \end{aligned}$$

Solution: Method 2—

$$\begin{aligned}\text{Slope of } PQ &= \frac{-1/3 - (-1/3 + r \sin \theta)}{-1 - (-1 + r \cos \theta)}, \\ &= \frac{-r \sin \theta}{-r \cos \theta}, \\ &= \tan \theta.\end{aligned}$$

$\therefore$ Equation of PQ : $y + 1/3 = \tan \theta(x + 1)$.

This intersects $y = 3x^2 + 2$,

so $3x^2 + 2 + 1/3 = \tan \theta(x + 1)$,

$$9x^2 - 3(\tan \theta)x - 3 \tan \theta + 7 = 0.$$

Substitute for Q , $x = -1 + r \cos \theta$:

$$9(-1 + r \cos \theta)^2 - 3 \tan \theta(-1 + r \cos \theta) - 3 \tan \theta + 7 = 0,$$

$$9(1 - 2r \cos \theta + r^2 \cos^2 \theta) + 3 \tan \theta - 3r \sin \theta - 3 \tan \theta + 7 = 0,$$

$$9 - 16r \cos \theta + 9r^2 \cos^2 \theta - 3r \sin \theta + 7 = 0,$$

$$9r^2 \cos^2 \theta - 18r \cos \theta - 3r \sin \theta + 16 = 0,$$

$$\text{i.e. } 9r^2 \cos^2 \theta - 3r(\sin \theta + 6 \cos \theta) + 16 = 0.$$

(ii) Using b(i), show that $AB^2 = \frac{(\sin \theta - 2 \cos \theta)(\sin \theta + 14 \cos \theta)}{9 \cos^4 \theta}$.

3

$$\begin{aligned}\text{Solution: } r_1 + r_2 &= \frac{3(\sin \theta + 6 \cos \theta)}{9 \cos^2 \theta}, \text{ and } r_1 r_2 = \frac{16}{9 \cos^2 \theta}, \\ AB^2 &= (r_2 - r_1)^2, \text{ (} = r_2^2 - 2r_2 r_1 + r_1^2 \text{)} \\ &= (r_2 + r_1)^2 - 4r_2 r_1, \\ &= \left(\frac{\sin \theta + 6 \cos \theta}{3 \cos^2 \theta} \right)^2 - 4 \times \frac{16}{9 \cos^2 \theta}, \\ &= \frac{\sin^2 \theta + 12 \sin \theta \cos \theta + 36 \cos^2 \theta - 64 \cos^2 \theta}{9 \cos^4 \theta}, \\ &= \frac{\sin^2 \theta + 12 \sin \theta \cos \theta - 28 \cos^2 \theta}{9 \cos^4 \theta}, \\ &= \frac{(\sin \theta - 2 \cos \theta)(\sin \theta + 14 \cos \theta)}{9 \cos^4 \theta}.\end{aligned}$$

(iii) Let L_1 be a tangent to the parabola $y = 3x^2 + 2$ from P , with point of contact R . Using the above results, find the two possible slopes of L_1 .

2

Solution: Method 1—

At the point of tangency, $AB^2 = 0$,

$$\text{i.e. } (\sin \theta - 2 \cos \theta)(\sin \theta + 14 \cos \theta) = 0.$$

$$\Rightarrow \sin \theta = 2 \cos \theta, \text{ or } \sin \theta = -14 \cos \theta,$$

$$\therefore \tan \theta = 2, \quad \tan \theta = -14.$$

Hence the slopes of L_1 are 2 or -14.

Solution: Method 2—

$$m_{PR} = \frac{y + 1/3}{x + 1} \text{ and for } \begin{cases} y = 3x^2 + 2, \\ y' = 6x. \end{cases}$$

$$\text{At the tangent points, } 6x = \frac{y + 1/3}{x + 1},$$

$$6x^2 + 6x - 1/3 = 3x^2 + 2,$$

$$9x^2 + 18x - 7 = 0,$$

$$(3x + 7)(3x - 1) = 0,$$

$$x = 1/3, -7/3.$$

$$\therefore y' = 6 \times 1/3, 6 \times (-7/3),$$

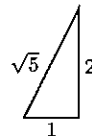
$$= 2, -14 \text{ (the slopes of } L_1 \text{)}.$$

(iv) Show that $PR = \frac{4\sqrt{5}}{3}$ when one of the slopes of L_1 has a value of 2.

Solution: Method 1—

When the slope of $L_1 = 2$, $\sin \theta = \frac{2}{\sqrt{5}}$, and $\cos \theta = \frac{1}{\sqrt{5}}$.

$\therefore R$ is at $\left(-1 + \frac{r}{\sqrt{5}}, -\frac{1}{3} + \frac{2r}{\sqrt{5}}\right)$



At the point of tangency to $y = 3x^2 + 2$,

$$\frac{dy}{dx} = 6x = 2,$$

$$\text{i.e. } x = \frac{1}{3}.$$

$$\text{But } \frac{1}{3} = -1 + \frac{r}{\sqrt{5}},$$

$$\frac{4}{3} = \frac{r}{\sqrt{5}},$$

$$r = \frac{4\sqrt{5}}{3},$$

$$\text{Thus } PR = \frac{4\sqrt{5}}{3}.$$

Solution: Method 2—

PR is a double root of $9r^2 \cos^2 \theta - 3r(\sin \theta + 6 \cos \theta) + 16 = 0$.

$$\text{So } PR^2 = \frac{16}{9 \cos^2 \theta},$$

$$PR = \sqrt{\frac{16}{9 \cos^2 \theta}},$$

$$= \frac{4}{3} \times \frac{1}{\cos \theta}.$$

But, as $\tan \theta = 2$,

$$\cos \theta = \frac{1}{\sqrt{5}},$$

$$\text{thus } PR = \frac{4\sqrt{5}}{3}.$$

Solution: Method 3—

Equation of PR : $y + \frac{1}{3} = 2(x + 1)$,

$$3y + 1 = 6x + 6,$$

$$6x - 3y + 5 = 0.$$

R is at the point of tangency to $y = 3x^2 + 2$,

$$\text{i.e. } 6x - 3(3x^2 + 2) + 5 = 0,$$

$$6x - 9x^2 - 6 + 5 = 0,$$

$$9x^2 - 6x + 1 = 0,$$

$$(3x - 1)^2 = 0,$$

$$\therefore x = \frac{1}{3}.$$

$$\text{But } \frac{1}{3} = -1 + \frac{r}{\sqrt{5}},$$

$$\frac{4}{3} = \frac{r}{\sqrt{5}},$$

$$r = \frac{4\sqrt{5}}{3},$$

$$\text{Thus } PR = \frac{4\sqrt{5}}{3}.$$

Solution: Method 4—

Equation of PR : $3y + 1 = 6x + 6$,

$$6x - 3y + 5 = 0.$$

R is at the point of tangency to $y = 3x^2 + 2$,

$$\text{i.e. } 6x - 3(3x^2 + 2) + 5 = 0,$$

$$6x - 9x^2 - 6 + 5 = 0,$$

$$9x^2 - 6x + 1 = 0,$$

$$(3x - 1)^2 = 0,$$

$$\therefore x = \frac{1}{3}.$$

Substitute in equation PR : $6 \times \frac{1}{3} - 3y + 5 = 0$,

$$-3y = -7,$$

$$y = \frac{7}{3}.$$

$$\begin{aligned}\text{So the length of } PR &= \sqrt{\left(-1 - \frac{1}{3}\right)^2 + \left(-\frac{1}{3} - \frac{7}{3}\right)^2}, \\ &= \sqrt{\frac{16}{9} + \frac{64}{9}}, \\ &= \sqrt{\frac{80}{9}}, \\ &= \frac{4\sqrt{5}}{3}.\end{aligned}$$